Project Design Phase

Solution Architecture

Date	02 October 2026
Team ID	SWTID-2026-5170
Project Name	Script-Controlled ACL – Restrict Record Access Based on Custom Script
Maximum Marks	4 Marks

Architecture diagram

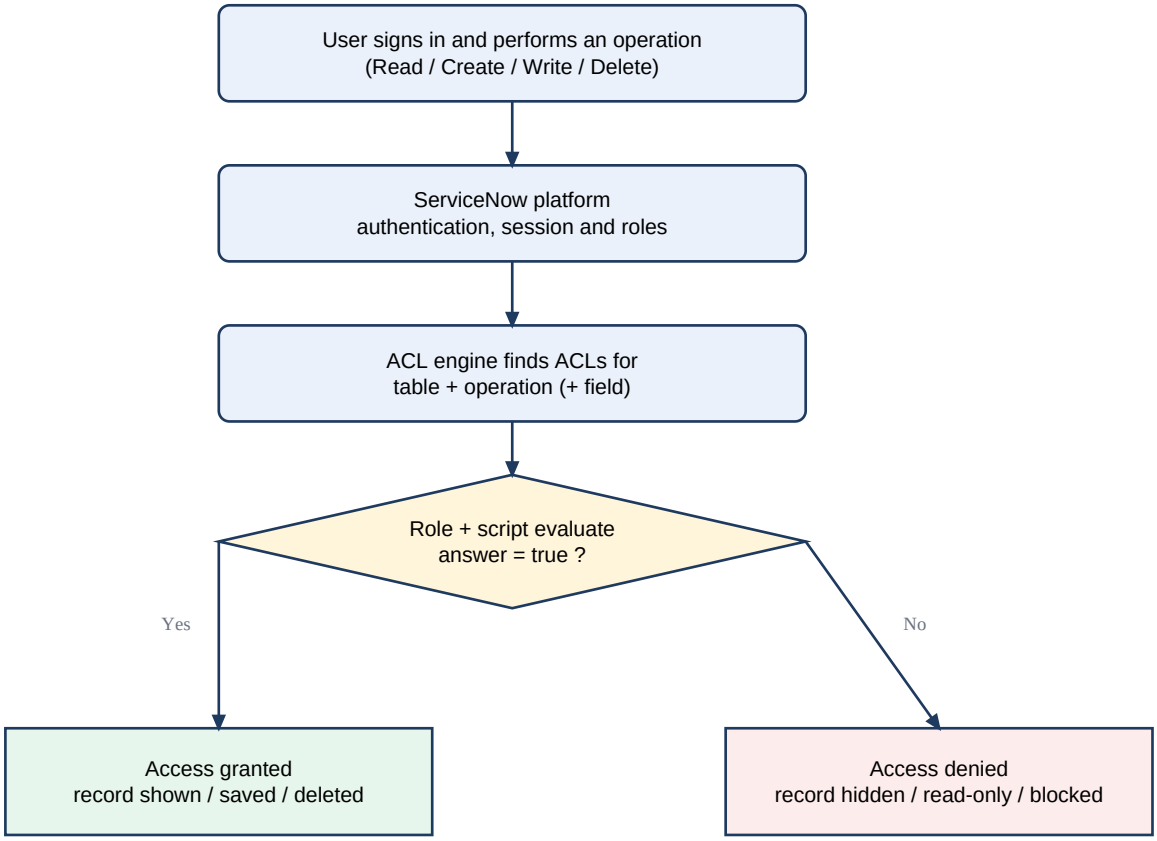

Figure 2: Script-controlled ACL decision flow

1. Technology stack

- **Core platform:** ServiceNow Personal Developer Instance.
- **Key modules:** User Administration, Tables and Columns, Access Control (ACL), Impersonate User.

2. Structure and behaviour

- **Trigger:** any request to read, create, write or delete a record in the table.
- **Evaluation:** an ACL grants access only when its role requirement, condition and script all evaluate to true.
- **Script:** uses `gs.hasRole()`, `gs.getUserID()` and `current.field` to decide.
- **Result:** access is granted or denied on the server.

3. Development phases

- Phase 1 - Users, roles and table creation (Milestones 1-2).

- Phase 2 - READ, CREATE, WRITE and DELETE ACL scripts (Milestones 3-6).
- Phase 3 - Testing with impersonation and documentation (Milestone 7).

4. Specifications

- **Inputs:** logged-in user, user roles, record field values.
- **Processing:** ACL script evaluation.
- **Outputs:** allow or deny for each operation.